



UNIT CODE: PRT582
SOFTWARE ENGINEERING: PROCESS AND TOOLS

Group Assignment: Software Project Plan
Warehouse Inventory System

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3. Background:

The idea of warehouse inventory control entails watching over and directing how materials move and are stored in a warehouse. This has to do with ensuring that incoming goods are unhampered during their move to the receiving area; finding and directing the movement of materials to their assigned locations; and ensuring that everything is picked and packed correctly for outbound shipments. Systems used may include barcode scanning, computerized inventory management, and even some automation.

4. Aims and Scope:

4.1. Aims:

Warehouse inventory - A web-based application where we manage inventory as the warehouse owner, also controlling categories, product details, sales, and reports. A completely free PHP backend with HTML, CSS, and JS frontend is used in this application. Below is the list of goals we are trying to achieve with this inventory system.

- **Inventory Management:**

It will be poised to handle product details and stock. With real-time data, it displays current stock levels.

- **User Role Management:**

With this system, different user roles can be created, like admin, regular user, and special user, and features can be assigned based on specific user role.

- **Sales Record:**

It will keep sales records that store product name, quantity, amount, and date of sale data. This data will be used for stock. Based on sales reports, user can analyze sales data and make decisions for their business.

- **User Experience:**

It is a responsive system built to deliver optimal user experiences on computers, tablets, and phones.

- **Speed and Performance:**

Optimized database design and query make this system faster, and it can handle an

extra load of traffic and product data.

- **Security:**

To keep data safe, all kinds of security measures are taken for a safe login experience, and md5 encryption is used to store passwords.

- **Compatibility:**

Finally, this system is developed for cross platforms so that all kinds of users can use this system.

4.2. Scope:

The project scopes are listed below:

- i. **Dashboard:**
 - It shows an overview of total users, categories, products, and sales.
- ii. **User Management:**
 - Add new users, and edit, or delete existing users.
 - User groups can be added, updated, and deleted as well.
- iii. **Product Management:**
 - New products can be added, existing products can be edited (categories and images can be assigned from categories and media sections), and deleted.
 - New products can be added; existing products can be edited (categories and images can be assigned from categories and media sections) and deleted.
 - New stocks can be added, and buying and selling prices can be updated here.
- iv. **Categories:**
 - For categorizing product, new categories can be added and existing categories can be modified and deleted.
- v. **Media Management:**
 - New media files can be uploaded, viewed, and deleted for products.
- vi. **Sales Management:**
 - Each sales record will be stored here. They can be modified and deleted.

vii. Sales Record:

- For a specific date range, daily sales and monthly sales records can be generated.

5. Requirements:

5.1. Functional Requirements:

The functional requirements for this project are given below:

User Group	Requirements
Admin	Add, update, and delete inventory items.
	Manage user roles and permissions.
	Generate detailed reports on inventory levels, sales, and orders.
	Monitor system activity and logs for security and auditing purposes.
	Configure system settings and preferences.
Warehouse Staff	Update inventory status (e.g., received, shipped, damaged)
	Manage and process orders (create, track, fulfill).
	View real-time stock levels and inventory status.
	Perform stock take and adjust inventory counts.
	Receive notifications about low stock levels and pending orders.
	Scan barcodes for quick inventory updates.
Guest	View the product catalogue and check stock availability
	Search for products using various filters (e.g., category, name, SKU).
	Request information about products and availability.
Delivery Staff	Access delivery schedules and routes.
	Update the status of deliveries (e.g., in transit, delivered).
	Confirm receipt of goods at delivery locations.
IT Support	Perform system maintenance and updates.
	Troubleshoot technical issues and resolve them quickly to ensure downtime remains at a minimum.
	Ensure regular data backups are performed, and restore operations work correctly.

Table 1: Functional Requirements

5.2. Non-Functional Requirements:

The non-functional requirements for this project are given below:

Security	Usability	Performance
Use strong encryption and secure logins to ensure no sensitive inventory data is stolen and user information is safe.	Make the interface easy to use and navigate, keeping it simple for administrators and staff to manage the system.	Handle more users and inventory items smoothly without slowing down.
Control user access tightly to prevent unauthorized people from accessing inventory data.	Make the design accessible to people with disabilities, following all necessary standards.	Make the system fast, allowing quick searches, updates, and order processing even with large inventories.
Encrypt all data sent between users and the server to stop anyone from intercepting it.	Make the interface straightforward with clear labels and instructions for all tasks.	Perform well even during busy times and when handling large amounts of data.
Audit regular security check to find and fix any vulnerabilities.	Develop the system in such way that should work consistently across different devices and screen sizes.	Needs to be reliable, with minimal downtime and high availability.
Backup the system regularly to avoid any loss.	Allow users to access training materials and give support to help them use all the features effectively.	Use resources efficiently to avoid slowdowns and crashes.
Meet all relevant data protection laws and standards.	Support multiple languages to cater to a diverse range of users.	Provide real-time updates and notifications to keep operations running smoothly
Secure any kind of document sharing process to prevent unauthorized access or leaks of sensitive information.	Make the system's collaboration features easy to understand, promoting teamwork among users.	Ensure all tasks, like inventory updates and order processing, are completed quickly without delays.

Table 2: Non-Functional Requirements

6. Agile Testing:

To ensure the Warehouse Inventory System performs reliably and satisfies user demands, agile testing is very crucial. The Agile Testing Quadrants will be used to organize and execute the testing strategy for this project. Several types of agile tests will be conducted which are described below in detail:

- **First Quadrant: Technology-Facing Tests that Support the Team (Automated):**

- i. **Unit Tests:**

- **Purpose:** To ensure that all the individual units of code (e.g., functions, methods) work correctly.
 - **Implementation:**
 - Use PHPUnit to write unit tests for the PHP backend functions.
 - Focus on critical functions such as user authentication, product management (add, edit, delete), and sales recording.
 - Verify that each function returns the expected results for a variety of input scenarios.

- ii. **Component Tests:**

- **Purpose:** To validate that each component of the system functions correctly on its own.
 - **Implementation:**
 - Test components such as the login system, user management, product management, and sales reporting.
 - Use Selenium to automate the testing of web interface components, ensuring they perform correctly across different browsers and devices.

- **Second Quadrant: Business-Facing Tests that Support the Team (Manual and Automated):**

- i. **Functional Tests:**

- **Purpose:** To confirm that the system performs as intended for all user roles.
 - **Implementation:**
 - Conduct manual testing to verify that admins, special users, and regular users can perform their respective tasks.
 - Automate functional tests where feasible to ensure repeatability and efficiency, using tools like Selenium.

- ii. **Story Tests:**

- **Purpose:** To ensure the system meets user requirements as defined by user stories.
 - **Implementation:**
 - Create test cases based on user stories, such as, " I want to add a new product as an admin so that I can manage the inventory."
 - Execute these tests manually to confirm that the system satisfies all the criteria.

- iii. **Prototypes and Simulations:**

- **Purpose:** To test the system's behavior in simulated real-world scenarios.
 - **Implementation:**
 - Create mock data to simulate typical usage patterns, such as high-volume product transactions or multiple users accessing the system simultaneously.
 - Conduct these tests manually and with automated scripts to assess system performance under realistic conditions.

- **Third Quadrant: Business-Facing Tests that Critique the Product (Manual):**

- i. **Exploratory Testing:**

- **Purpose:** To find out any significant defects that might be missed in structured testing.
 - **Implementation:**
 - Perform exploratory testing sessions where testers navigate the system freely, looking for unexpected behavior or defects.
 - Record any issues discovered and retest after fixes are applied.

- ii. **Scenarios and Usability Testing:**

- **Purpose:** To make sure the system being user-friendly and satisfies business requirements.
 - **Implementation:**
 - Test several user situations to ensure the system is intuitive and simple.
 - Organize usability testing sessions with actual users to learn about their feedbacks on the system's design and operation.

- iii. **User Acceptance Testing (UAT):**

- **Purpose:** To ensure the system meets both the needs and expectations of the end users.
 - **Implementation:**
 - Engage users to test critical capabilities to ensure the system meets their needs.
 - Gather feedback from users and make adjustments as required.

- **Fourth Quadrant: Technology-Facing Tests that Critique the Product (Automated and Tools):**

- i. **Performance and Load Testing:**

- **Purpose:** To ensure that the system performs well under expected load conditions.

- **Implementation:**

- To simulate several users accessing the system at once, utilize Apache JMeter.
- Track out performance bottlenecks and analyze response times.

ii. **Security Testing:**

- **Purpose:** To identify and address vulnerabilities in the system.

- **Implementation:**

- Conduct penetration testing to search for common security vulnerabilities like SQL injection, cross-site scripting, and cross-site request forgery (CSRF) (XSS).
- Guarantee the security of user data transfer and storage.

iii. **"ility" Testing (Scalability, Reliability, etc.):**

- **Purpose:** To test the system's non-functional features, such as dependability and scalability.
- **Implementation:**
 - To assess scalability, simulate growing user loads and data quantities.
 - Confirm that the system is dependable and accessible in a variety of scenarios, including those involving server outages or network disturbances.

We will make sure that our Warehouse Inventory System is fully tested and prepared for deployment by adhering to the Agile Testing Quadrants. We will be able to detect and resolve any problems early on with the aid of our thorough testing approach, guaranteeing a dependable and seamless user experience.

7. Conclusion:

The efficient administration of store inventory is the keystone upon which the smooth operation of business rests. For one, it guarantees the ready availability of product right when the customer wants it, thereby obviating the proper loss of any customer due to a stock-out of any pertinent product. For two, in what is essentially the most labor- and time-consuming part

of the business (the picking-and-packing operation), technologically advanced inventory systems speed up order fulfillment and boost productivity. Ultimately, they also boost the business's bottom line. For those reasons, having a proper inventory system in place is crucial to any business's operational excellence.